Near-optimal nonlinear erasure list-decodable codes via one-bit strong dispersers and an unbalanced two-source extractor

Avraham Ben-Aroya, Dean Doron, Amnon Ta-Shma

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Editorial note. Editable transcription from the official CC BY3.0 Unported publication, not relicensed4.0. Full core author proof retained: Sections2–4 plus5.1/5.3; historical comparison tables, nonstrong sketch5.2, Ramsey applications5.4 and open questions omitted. Main statement is originalTheorem2; detailedTheorems27/40/46 and all source-local supporting proofs remain in one item. Original formulas/inequalities, including Definition24 notation and Lemma44 strictness factors, are transcribed without mathematical repair or added assumptions. Only line wrapping and presentation macros adapted. Original numbered bracket references retain complete45-entry ordinary bibliography. All production/validation on Yupeng; root owns compilation.

Allowed standard prerequisites (editorial).

- Statistical distance and min-entropy, Raz strong two-source extractor, curve mergers, correlation breakers with advice, resilient functions. OriginalSection2 retains exact source-used definitions and cited statements; published standard components are cited rather than recursively reproved

Main guarantee and conventions (original Sections 1–2)

A code $\mathcal{C} \subseteq \{0,1\}^{\bar{n}}$ is (s, L) erasure list-decodable if for every word w, after erasing any s symbols of w, the remaining $\bar{n} - s$ symbols have at most L possible completions into a codeword of $\mathcal{C}$.

Theorem 2 (see also Theorem 46)**.** *For every constant $0 < \gamma < 1$ and $\varepsilon = n^{-\Omega_\gamma(1)}$ there exists an explicit $((1-\varepsilon)\bar{n}, L = \log^{O_\gamma(1)} \frac{1}{\varepsilon})$ erasure list-decodable code $\mathcal{C} : \{0,1\}^n \to \{0,1\}^{\bar{n}}$ of rate $\varepsilon^{1+\gamma}$.*

We stress that the codes we present are explicit in the sense that they have explicit encoding, but we do not know whether the codes we construct admit efficient erasure list-decoding algorithms. We also mention that the list-size $\mathrm{poly}(\log \frac{1}{\varepsilon})$ achieved by our code is exponentially smaller than the best possible list-size by any linear code. Throughout the paper we use the convention that

lowercase variables are the logarithm (in base 2) of their corresponding uppercase variables, e.g., $n = \log N$, $d = \log D$. We denote by $[A]$ the set $\{1, \ldots, A\}$. The density of a set $B \subseteq A$ is $\rho(B) = |B|/|A|$. We say a function $f : A \to B$ is explicit if there exists a deterministic polynomial algorithm that runs in time $\mathrm{poly}(\log |A|)$ and computes f.

Preliminaries (original Section 2)

Random variables and min-entropy

The statistical distance between two distributions X and Y on the same domain Ω is defined as $|X - Y| = \max_{A \subseteq \Omega}(\Pr[X \in A] - \Pr[Y \in A])$. If $|X - Y| \leq \varepsilon$ we say X is ε-close to Y and denote it by $X \approx_\varepsilon Y$. We denote by U_n the random variable distributed uniformly over $\{0,1\}^n$. We say a random variable is flat if it is uniform over its support. For a function $f : \Omega_1 \to \Omega_2$ and a random variable X distributed over Ω_1, $f(X)$ is the random variable distributed over Ω_2 obtained by choosing x according to X and computing $f(x)$. For a set $A \subseteq \Omega_1$, $f(A) = \{f(x) \mid x \in A\}$. For every $f : \Omega_1 \to \Omega_2$ and two random variables X and Y distributed over Ω_1, it holds that $|f(X) - f(Y)| \leq |X - Y|$. The min-entropy of a random variable X is defined by

$$H_\infty(X) = \min_{x \in \mathrm{Supp}(X)} \log \frac{1}{\Pr[X = x]}.$$

A random variable X is an (n, k) source if X is distributed over $\{0,1\}^n$ and has min-entropy at least k. When n is clear from the context we sometimes omit it and simply say that X is a k-source. Every k-source X can be expressed as a convex combination of flat distributions each with min-entropy at least k.

Definition 6 (average conditional min-entropy)**.** *Let X, Y be two random variables. The average conditional min-entropy of X given Y is*

$$\widetilde{H}_\infty(X \mid Y) = -\log\left(\mathbb{E}_{y \sim Y}\left[2^{-H_\infty(X|Y=y)}\right]\right).$$

We will use the following simple claim about average conditional min-entropy:

Claim 7. *For any random variables X, Y, $\widetilde{H}_\infty(X \mid Y) \geq H_\infty(X) - \log|\mathrm{Supp}(Y)|$.*

Condensers

Definition 8 (condenser)**.** *A function $C : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ is an $(n,k) \to_{\varepsilon_{\mathrm{cond}}} (m,k')$ condenser, if for every (n,k) source X, $C(X, U_d)$ is $\varepsilon_{\mathrm{cond}}$-close to an (m,k') source. If $k = \delta n$ and $k' = \delta' m$ we say C is a $\delta \to_{\varepsilon_{\mathrm{cond}}} \delta'$ condenser.*

Lemma 9. *Suppose $C : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ is an $(n,k) \to_{\varepsilon_{\mathrm{cond}}} (m, k' + d)$ condenser. Let X be an (n,k) source. Let ε_i be the minimal distance of $C(X,i)$ to an (m,k') source. Then, $\mathbb{E}_{i \in \{0,1\}^d}[\varepsilon_i] \leq \varepsilon_{\mathrm{cond}}$.*

Proof. Fix an (n,k) source X. For $i \in \{0,1\}^d$, let $H_i \subseteq \{0,1\}^m$ be the set of "heavy" elements of $C(X,i)$,

$$H_i = \{w \in \{0,1\}^m : \Pr_{x \in X}[C(x,i) = w] \geq 2^{-k'}\}.$$

The distance of $C(X,i)$ from a k'-source is

$$\varepsilon_i = \Pr_{x\in X}[C(x,i)\in H_i] - 2^{-k'}|H_i|,$$

by redistributing the mass of the heavy elements. Let $H = \bigcup_{i\in\{0,1\}^d} H_i$. Then,

- For every $w \in H$, $\Pr_{x\in X, i\in\{0,1\}^d}[C(x,i)=w] \geq 2^{-d}2^{-k'} = 2^{-(k'+d)}$, and,
- it holds that

$$\begin{aligned}
\varepsilon_{\text{cond}} &= \Pr_{x\in X, i\in\{0,1\}^d}[C(x,i)\in H] - |H|2^{-(k'+d)} \\
&= \sum_{i\in\{0,1\}^d} 2^{-d}\Pr_x[C(x,i)\in H] - |H|2^{-(k'+d)} \\
&\geq \sum_{i\in\{0,1\}^d} 2^{-d}\Pr_x[C(x,i)\in H_i] - 2^{-(k'+d)}\sum_{i\in\{0,1\}^d}|H_i| \\
&= \sum_{i\in\{0,1\}^d} 2^{-d}(\Pr_x[C(x,i)\in H_i] - 2^{-k'}|H_i|) \\
&= \sum_{i\in\{0,1\}^d} 2^{-d}\varepsilon_i = \mathbb{E}_{i\in\{0,1\}^d}[\varepsilon_i].
\end{aligned}$$

□

Two-source extractors

Definition 10 (two-source extractor)**.** *A function* $\mathsf{2Ext} : \{0,1\}^{n_1}\times\{0,1\}^{n_2}\to\{0,1\}^m$ *is an* $((n_1,k_1),(n_2,k_2),\varepsilon)$ *two-source extractor if for every two independent sources* X_1 *and* X_2 *where* X_1 *is an* (n_1,k_1) *source and* X_2 *is an* (n_2,k_2) *source, it holds that* $\mathsf{2Ext}(X_1,X_2)\approx_\varepsilon U_m$*. We say that* $\mathsf{2Ext}$ *is strong if*

$$(\mathsf{2Ext}(X_1,X_2),X_1)\approx_\varepsilon (U_m,X_1) \quad \textit{and} \quad (\mathsf{2Ext}(X_1,X_2),X_2)\approx_\varepsilon (U_m,X_2).$$

In our construction, we will use the following two-source extractor:

Theorem 11 ([39])**.** *For every constant* $\delta_{\text{Raz}} > \frac{1}{2}$ *there exist constants* $c_1 = c_1(\delta_{\text{Raz}}), c_2 = c_2(\delta_{\text{Raz}}) > 1$ *such that for every* n_1,k_1,n_2,k_2 *satisfying*

$$k_1 \geq c_1\log n_2, \qquad k_2 \geq c_2\log n_1,$$

there exists an explicit strong $((n_1,k_1),(n_2,k_2=\delta_{\text{Raz}}n_2),\varepsilon_{\text{Raz}})$ *two-source extractor*

$$\text{Raz} : \{0,1\}^{n_1}\times\{0,1\}^{n_2}\to\{0,1\}^m$$

with $m = \Omega(\min\{k_1,k_2\})$ *and* $\varepsilon_{\text{Raz}} = 2^{-\Omega(m)}$*, where the constants hiding in the asymptotic notation may depend on* δ_{Raz}*.*

Claim 12. *Suppose* $\mathsf{2Ext} : \{0,1\}^{n_1}\times\{0,1\}^{n_2}\to\{0,1\}^m$ *is a strong* $((n_1,k_1),(n_2,k_2),\varepsilon)$ *two-source extractor. Let* X *be an* (n,k_1) *source. Call an element* $y\in\{0,1\}^{n_2}$ *bad if* $|\mathsf{2Ext}(X,y)-U_m| > \varepsilon$*, and let* B_Y *denote the set of all bad elements. Then,* $|B_Y| < 2^{k_2}$*.*

Proof. Assume towards contradiction that $|B_Y| \geq 2^{k_2}$ and let Y be the uniform distribution over the set B_Y. Then, $H_\infty(Y) \geq k_2$ and so $(\mathsf{2Ext}(X,Y),Y) \approx_\varepsilon (U_m,Y)$ which implies that

$$\frac{1}{|B_Y|}\sum_{y\in B_Y}|\mathsf{2Ext}(X,y)-U_m| \leq \varepsilon.$$

However, $|\mathsf{2Ext}(X,y)-U_m| > \varepsilon$ for every $y \in B_Y$, in contradiction. □

Mergers

A merger takes as input a list of possibly correlated random variables along with a short uniform seed and outputs one random variable which is close to having high min-entropy, provided at least one of the input variables has high min-entropy. Formally:

Definition 13 (somewhere-random source)**.** *A source $X = X_1 \circ \ldots \circ X_A$ is an $(n,k,(\alpha,\beta))$ somewhere-random (s.r.) source if there is a random variable $I \in \{0,\ldots,A\}$ such that for every $i \in [A]$, $(X_i \mid I=i)$ is α-close to an (n,k) source and $\Pr[I=0] \leq \beta$. The variable I is called the indicator of source. If $\alpha=\beta=0$ we say X is a (n,k) s.r. source.*

Definition 14 (merger)**.** *A function $B : (\{0,1\}^n)^D \times \{0,1\}^t \to \{0,1\}^m$ is a (k,k',ε) merger, if for every (n,k) s.r. source $X = X_1 \circ \ldots \circ X_A$, the output $M(X,U_t)$ is ε-close to a k'-source.*

There are explicit constructions of good mergers. Dvir and Wigderson [17] constructed the curve merger and proved that it works with $t = O(\log \frac{n}{\varepsilon})$. This was further improved in [16] who proved that $t = O(\log \frac{D}{\varepsilon})$ suffices. Notice that now t only depends on the number of sources D and the requested error ε, but not on the source length n, and this remarkable property will be crucial for us. Formally,

Theorem 15 ([17,16])**.** *There exists a constant $c_{\mathrm{DKSS}} \geq 1$ such that the following holds. Fix $\beta,\delta,\varepsilon > 0$. There exists an explicit function $B : (\{0,1\}^n)^D \times \{0,1\}^t \to \{0,1\}^n$ that is a $(k=\delta n, k'=(1-\beta)\delta n, \varepsilon)$ merger, with $t = c_{\mathrm{DKSS}} \cdot \frac{1}{\beta}\log\frac{D}{\varepsilon}$.*

Correlation breakers with advice

A correlation-breaker with advice is a function $\mathrm{CBA} : \{0,1\}^n \times \{0,1\}^\ell \times [A] \to \{0,1\}^m$ where we think of the first input as a weak source, the second as an independent short seed and the last as an advice string. Roughly speaking, applying CBA on t possibly correlated seeds with t distinct advice strings results in independent random variables. For example, $\mathrm{CBA}(X,Y,\alpha)$ is (nearly) independent of $\mathrm{CBA}(X,Y,\alpha')$ for any $\alpha \neq \alpha'$. Formally,

Definition 16. *A function $\mathrm{CBA} : \{0,1\}^n \times \{0,1\}^\ell \times [A] \to \{0,1\}^m$ is a $(t,k,\varepsilon_{\mathrm{CBA}})$ correlation-breaker with advice if the following holds. If Y is a distribution over $\{0,1\}^n$, $Z = (Z_1,\ldots,Z_t)$ is a distribution on $(\{0,1\}^\ell)^t$, H is a random variable and $\delta > 0$ satisfy:*

- *Y and Z are independent, conditioned on H,*

- $\widetilde{H}_\infty(Y \mid H) \geq k + \log(1/\varepsilon_{\mathrm{CBA}})$,
- $(Z_1, H) \approx_\delta (U_\ell, H)$, *and,*
- $\alpha_1, \ldots, \alpha_t \in [A]$ *are distinct strings.*

Then,

$$(\mathrm{CBA}(Y, Z_1, \alpha_1), (\mathrm{CBA}(Y, Z_i, \alpha_i))_{i=2}^t, H) \approx_{\delta + 2\varepsilon_{\mathrm{CBA}}} (U_m, (\mathrm{CBA}(Y, Z_i, \alpha_i))_{i=2}^t, H).$$

We use the following result:

Theorem 17 ([13, Theorem 4.12])**.** *There exists a constant $c_{\mathrm{CBA}} \geq 1$ such that the following holds. Let n, a be integers and $\varepsilon_{\mathrm{CBA}} > 0$. Then, there exists an explicit $(t, k_{\mathrm{CBA}}, \varepsilon_{\mathrm{CBA}})$ correlation-breaker with advice*

$$\mathrm{CBA} : \{0,1\}^n \times \{0,1\}^\ell \times [A] \to \{0,1\}$$

with $\ell = c_{\mathrm{CBA}} \cdot at \cdot \log \frac{n}{\varepsilon_{\mathrm{CBA}}}$ and $k_{\mathrm{CBA}} \geq \ell$.

In our setting, the number of rows A is a constant independent of n. For this reason we work with a "basic" correlation-breaker, where there is no attempt to optimize the dependence of ℓ on a. This gives a seed-length which is optimal up to constant multiplicative factors. We also need the following lemma.

Lemma 18. *Let $X_1, \ldots, X_t$ be random variables over $\{0,1\}^m$. Further suppose that for any $i \in [t]$,*

$$(X_i, \{X_j\}_{j \neq i}) \approx_\varepsilon (U_m, \{X_j\}_{j \neq i}).$$

Then, $(X_1, \ldots, X_t) \approx_{t\varepsilon} U_{tm}$.

Limited independence and non-oblivious bit-fixing sources

Definition 19. *A distribution X over $\{0,1\}^A$ is called (t, γ)-wise independent if the restriction of X to every t coordinates is γ-close to U_t. A source X over $\{0,1\}^A$ is called a (q, t, γ) non-oblivious bit-fixing source if there exists a subset $Q \subseteq A$ of size at most q such that the joint distribution of the bits in $A \setminus Q$ is (t, γ)-wise independent. The bits in Q are allowed to arbitrarily depend on the bits in $A \setminus Q$. If $\gamma = 0$ we often say that X is a (q, t) non-oblivious bit-fixing source.*

Lemma 20 ([3])**.** *A (t, γ)-wise distribution over A bits is $(A^t\gamma)$-close to some t-wise independent distribution.*

Definition 21. *Let $f : \{0,1\}^A \to \{0,1\}$, $\mathcal{D}$ a distribution over $\{0,1\}^A$ and $Q \subseteq A$. Let $I_{Q,\mathcal{D}}(f)$ denote the probability that f is undetermined when the variables outside Q are sampled from $\mathcal{D}$. We define $I_{q,t,\gamma}(f)$ to be the maximum of $I_{Q,\mathcal{D}}(f)$ over all $Q \subseteq A$ of size q and all $\mathcal{D}$ that is a (t, γ)-wise independent distribution. We say that f is (t, γ)-independent (q, ε)-resilient if $I_{q,t,\gamma}(f) \leq \varepsilon$.*

Theorem 22 ([9,32])**.** *For every $0 < \gamma < 1$ there exists a constant $c_\gamma \geq 1$ such that for all $A > 0$ there exists an explicit function $f : \{0,1\}^A \to \{0,1\}$ with the following property: For every $t \geq c_\gamma \log^4 A$,*

- *f is almost balanced: For any t-wise independent distribution $\mathcal{D}$ on $\{0,1\}^A$, $\Pr_{x\sim\mathcal{D}}[f(x)=1] = 1/2 \pm A^{-1/c_\gamma}$, and,*
- *f is resilient: $I_{q,t,\gamma}(f) \leq c_\gamma \cdot q/A^{1-\gamma}$.*

Constant Degree Condensers (original Section 3)

In this section we prove:

Theorem 23. *For every constant $0 < \delta_1 < \delta_2 = 0.7$, every $s \geq s_0(\delta_1)$ and every integer n_1 and $\varepsilon_{\text{cond}} \geq 2^{-\Omega(n_1)}$ there exists an explicit $\delta_1 \rightarrow_{\varepsilon_{\text{cond}}} \delta_2$ condenser $C : \{0,1\}^{n_1} \times \{0,1\}^d \to \{0,1\}^{n_2}$ with $n_2 = (\frac{2}{3})^s n_1$ and $d = 4c_{\text{DKSS}}(s + \log \frac{1}{\varepsilon_{\text{cond}}})$, where c_{DKSS} is the constant from Theorem 15. Note that d is independent of n_1.*

Note that, in particular, for every $\delta_1 > 0$ there exists an explicit $\delta_1 \rightarrow_{\varepsilon_{\text{cond}}} \delta_2 = 0.7$ condenser $C : \{0,1\}^{n_1} \times \{0,1\}^d \to \{0,1\}^{n_2}$ with $n_2 = \Omega(n_1)$ and $d = O(\log \frac{1}{\varepsilon_{\text{cond}}})$. However, we will need the more precise version that appears in Theorem 23. The proof goes through somewhere-random condensers, so let us first discuss the similarities and differences between condensers and somewhere-random condensers. We begin with the necessary definitions:

Definition 24 (s.r. condenser)**.** *A function $C : \{0,1\}^n \to (\{0,1\}^m)^A$ is an $(n,k) \rightarrow_\varepsilon (m,k')$ s.r. condenser if for every (n,k) source X it holds that $C(X) = C(X,1) \circ \ldots \circ C(X,A)$ is ε-close to a k' s.r. source. If $k = \delta n$ and $k' = \delta m$ we say C is $\delta \rightarrow_\varepsilon \delta'$ s.r. condenser.*

We may take a condenser $C : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ and expand it to a table with the outputs of all possible seeds, i.e., define $S : \{0,1\}^n \to (\{0,1\}^m)^D$, with $D = 2^d$, where $S(x)_i = C(x,i)$. The condenser property guarantees that for every k-source X, most rows in the table are close to having k' min-entropy. In contrast, a s.r. condenser is a weaker object, because it only guarantees that one row has k' entropy (or more precisely that we are in a convex combination of such cases). The major question we consider now is the dependence of the degree (2^d for condensers and A for s.r. condensers) on n, m, k, k' and ε. We focus on the case where $m = \Omega(n)$, $k = \delta n$, $k' = \delta' m$ and $\delta < \delta'$ are constants. A priori, we could have expected the degree to depend on n and ε, as is indeed the case when m might be arbitrarily small. However, remarkably, things are drastically different when $m = \Omega(n)$. In this case both condensers and s.r. condensers may be of degree that is independent of n and this will be crucial for us. If we consider the dependence on the error, then s.r. condensers may have exponentially-small error and constant D, whereas the degree of a condenser is at least $d \geq \log(\frac{1}{\varepsilon})$. Remarkably, all of that can be explicitly achieved, as we now explain. The basic building block we use is the following beautiful result of Zuckerman, which is based on additive combinatorics:

Theorem 25 ([45, Theorem 8.3])**.** *For every constant $0 < c < 1$ there exists a constant $\alpha = \alpha(c)$ such that for every constant $\delta \leq c$ and integer n there exists an explicit function $C : \{0,1\}^n \to (\{0,1\}^{\frac{2}{3}n})^2$ that is a $\delta \rightarrow_\varepsilon (1+\alpha)\delta$ s.r. condenser with $\varepsilon = 2^{-\Omega(\alpha\delta n)}$.*

Somewhere-random condensers can be easily composed. Specifically, Barak et al. [5] showed that if $C_1 : \{0,1\}^{n_1} \to (\{0,1\}^{n_2})^{\ell_1}$ is a $\delta_1 \to_{\varepsilon_1} \delta_2$ s.r. condenser and $C_2 : \{0,1\}^{n_2} \to (\{0,1\}^{n_3})^{\ell_2}$ a $\delta_2 \to_{\varepsilon_2} \delta_3$ s.r. condenser then $C_2 \circ C_1 : \{0,1\}^{n_1} \to (\{0,1\}^{n_3})^{\ell_1 \cdot \ell_2}$ defined by $C_2 \circ C_1(x)_{(i_1,i_2)} = C_2(C_1(x)_{i_1})_{i_2}$ is a $\delta_1 \to_{\varepsilon_1+\varepsilon_2} \delta_3$ s.r. condenser. Composing the s.r condenser of Theorem 25 with itself s times we get an explicit function $C : \{0,1\}^n \to (\{0,1\}^m)^D$ with $D = 2^s$ and $m = (\frac{2}{3})^s n$ that is a $\delta \to_\varepsilon \delta'$ s.r. condenser with $\varepsilon = \sum_{i=1}^{s} 2^{-\Omega((1+\alpha)^i \delta (\frac{2}{3})^i n)} = 2^{-\Omega(m)}$ and $\delta' \geq (1+\alpha(\delta'))^s \delta$. Therefore:

Lemma 26. *For every constants $0 < \delta_1 < \delta_2 < 1$ there exists a constant $s = s(\delta_1, \delta_2)$ and an explicit function $C : \{0,1\}^{n_1} \to (\{0,1\}^{n_2})^D$ that is a $\delta_1 \to_\varepsilon \delta_2$ s.r. condenser with $D = 2^s$, $n_2 = (\frac{2}{3})^s n_1$ and $\varepsilon = 2^{-\Omega(n_2)}$. Note that D is independent of n and ε.*

Right now, if X is a k-source, the table $C(X)$ has D rows, and, roughly speaking, the guarantee is that one of these rows has density δ'. We want to change this to get a condenser, i.e., we are willing to invest a short seed (that is independent of n) and we want to get one output which is close to uniform. (Alternatively, we can write the condenser as a table with one row per seed, the number of rows is independent of n and most rows are close to uniform.) This is exactly what a merger does and applying the merger of Theorem 15 with $\beta = \frac{1}{4}$ on the s.r. condenser of Lemma 26 (with δ_2 close to 1) gives Theorem 23.

The Unbalanced Two-Source Extractor Construction (original Section 4)

The main result of this section is the following two-source extractor.

Theorem 27. *For every integer n and two constants $\delta_0, \varepsilon_0 > 0$ there exists a constant c such that for $d \geq c \log n$ and $k \geq c \log d$ there exists an explicit $((n,k),(d,\delta_0 d),\varepsilon_0)$ two-source extractor $\mathsf{2Ext} : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}$.*

The extractor in the above theorem has constant error, and works when:

1. Each source's entropy is in the order of the logarithm of the length of the other source.
2. The shorter source, of length d, has an arbitrarily small constant density δ_0.

We think of n and $d = d(n)$ as growing parameters while ε_0 and δ_0 are constants. We use asymptotic notations (such as $\Omega(\cdot)$) to hide constants that are independent of n and d (but may depend on ε_0 and δ_0).

4.1 The Construction

Recall that ε_0 is the target error of the extractor $\mathsf{2Ext}$. The input to $\mathsf{2Ext}$ is a pair (x,y) where x is drawn from an (n,k) source X, and y is drawn from an independent $(d,\delta_0 d)$ source Y. Our problem is that the y comes from a $\delta_0 d$-source for some $\delta_0 < \frac{1}{2}$. To overcome this, we do the following:

- We apply the condenser of Theorem 23 on y to get a table y' that is 1-wise 0.7-dense. Notice that the output of this step is a table rather than a single output.
- We apply Raz's extractor (Theorem 11) on the table and the input x from the other source to convert the table y' to another table y'' that is 1-wise uniform.
- We apply the t correlation-breaker with advice of Theorem 17 on y, using the table y'' as the seed, to get a table y''' that is t-wise uniform.
- Finally, we apply the resilient function f of Theorem 22 on the table y''' to collapse the many rows of the table to a single, close to uniform, output.

Formally, these steps work as follows:

Condense the short source. We are given $\delta_0 < \frac{1}{2}$. Set $\delta' = 0.69$ and $\delta_2 = 0.7$. By Theorem 23 there exists a constant $s_0 = s_0(\delta_0)$ such that for every $s \geq s_0$ there exists an explicit

$$C : \{0,1\}^d \times \{0,1\}^a \to \{0,1\}^{d'}$$

that is a $\delta_0 \to_{\varepsilon_{\text{cond}}} \delta_2 = 0.7$ condenser with $a = 4c_{\text{DKSS}}(s + \log \frac{1}{\varepsilon_{\text{cond}}})$ and $d' = (\frac{2}{3})^s d$. We set

$$\gamma = \frac{1}{2^5 c_{\text{DKSS}}},$$

and this also fixes c_γ as in Theorem 22. Notice that γ and c_γ are fixed constants independent of all other parameters in our system. Now, choose $\varepsilon_{\text{cond}}$ so that

$$\left(\frac{1}{\varepsilon_{\text{cond}}}\right)^{\log(3/2)} \geq \frac{4}{\delta_0} 2^{12} c_\gamma c_{\text{DKSS}}^4 \log^4 \frac{1}{\varepsilon_{\text{cond}}}, \tag{1}$$

and also so that $\varepsilon_{\text{cond}} \leq \xi(\varepsilon_0, \delta_0)$, where

$$\xi(\varepsilon_0, \delta_0) = \min\left\{2^{-s_0}, \left(\frac{\varepsilon_0}{8}\right)^2, \left(\frac{\varepsilon_0}{5}\right)^{c_\gamma}, \left(\frac{\varepsilon_0}{5c_\gamma}\right)^{1/\gamma}\right\}. \tag{2}$$

Given $\varepsilon_{\text{cond}}$, we set $s = \log \frac{1}{\varepsilon_{\text{cond}}} \geq s_0$, giving $a = 8c_{\text{DKSS}} \log \frac{1}{\varepsilon_{\text{cond}}}$. Note that the degree of the condenser, $A = 2^a$, satisfies

$$\sqrt{\varepsilon_{\text{cond}}} A = 2^{-\frac{1}{2}\log \frac{1}{\varepsilon_{\text{cond}}} + a} = 2^{-\frac{a}{2^4 c_{\text{DKSS}}} + a} = A^{1-2\gamma}.$$

Observe that $s \geq s_0$ and that $d' = \Omega(d)$. Also, notice that $(\delta_2 - \delta')d' = d'/100 \geq a = \log A$ for large enough d. Thus, C is a $(d, \delta d) \to_{\varepsilon_{\text{cond}}} (d', \log(A) + \delta' d')$ condenser. Define an $A \times d'$ table Y' where

$$Y'_i = C(Y, i) \in \{0,1\}^{d'} \quad \text{for } i = 1, \ldots, A.$$

1-wise uniformity. Let c_1, c_2 be the constants from Theorem 11 for $\delta_{\mathrm{Raz}} = 0.6$. Notice that $\delta_{\mathrm{Raz}} d' = \Omega(d') = \Omega(d)$. Therefore, for a constant c large enough, $d \geq c \log n$ is large enough so that $\delta_{\mathrm{Raz}} d' \geq c_2 \log n$. We can, in particular, choose c such that in addition $c \geq c_1$. Recalling that $k \geq c \log d$, we have $k \geq c_1 \log d'$. By Theorem 11, there exists an explicit function

$$\mathrm{Raz} : \{0,1\}^n \times \{0,1\}^{d'} \to \{0,1\}^{d''}$$

that is a strong $((n,k),(d',\delta_{\mathrm{Raz}} d'),\varepsilon_{\mathrm{Raz}} = 2^{-\Omega(d'')})$ two-source extractor with $d'' = \Omega(\min\{k, \delta_{\mathrm{Raz}} d'\}) = \Omega(k)$.[1] Define an $A \times d''$ table Y'' where

$$Y''_i = \mathrm{Raz}(X, Y'_i) \quad \text{for } i = 1, \ldots, A.$$

t-wise uniformity. Let $k_{\mathrm{CBA}} = \frac{\delta_0 d}{8}$ and $\varepsilon_{\mathrm{CBA}} = \frac{1}{d}$. Set

$$t = \frac{\delta_0}{4} \left(\frac{3}{2}\right)^s.$$

Notice that for a large enough constant c we have $d'' = \Omega(k) = \Omega(c \log d) \geq c_{\mathrm{CBA}} a t \log \frac{d}{\varepsilon_{\mathrm{CBA}}}$, where the latter is the seed-length required by the correlation-breaker from Theorem 17. Also, $k_{\mathrm{CBA}} = \frac{\delta_0 d}{8} \geq d''$ for large enough d, as $d'' = \Omega(k) = \Omega(\log d)$. Hence, by Theorem 17 there exists an explicit function

$$\mathrm{CBA} : \{0,1\}^d \times \{0,1\}^{d''} \to \{0,1\}$$

that is a $(t, k_{\mathrm{CBA}}, \varepsilon_{\mathrm{CBA}})$ correlation-breaker with advice. Define an $A \times 1$ table Y''' where

$$Y'''_i = \mathrm{CBA}(Y, Y''_i, i) \quad \text{for } i = 1, \ldots, A.$$

Keep in mind that the entropy in Y suffices for CBA since $H_\infty(Y) = 8k_{\mathrm{CBA}}$.

Collapse. Take $f : \{0,1\}^A \to \{0,1\}$ to be the $(q = A^{1-2\gamma}, t, \varepsilon_f = c_\gamma A^{-\gamma})$ resilient function of Theorem 22 and output $f(y'''_1, \ldots, y'''_A)$.

4.2 Two Subtleties

As mentioned in the introduction, there are several delicate issues in the analysis:

1. Circular dependence: Y'' depends on both X and Y, and is used as a seed in the application of the correlation-breaker with advice on Y.
2. We need Y''' to be close to a perfect t-wise independent table, while the correlation-breaker with advice only guarantees that every t good rows are close to uniform. To bridge the gap we need the error to be at least polynomially-small in the number of rows, but some of the steps incur a large constant error.

To overcome the first issue we show a conditioning under which Y'' is still good, Y is independent of Y'' and even after the conditioning the two sources have enough min-entropy. To overcome the second issue we distinguish between large errors that depend on the number of rows A, and small errors that depend on the row length (see Section 1.3 in the introduction of the original publication). In particular, the errors are of three types:

[1] Although $k \geq c \log d$ we can always assume w.l.o.g. that $k = c \log d$ and so $k \leq \delta_{\mathrm{Raz}} d' = \Omega(d)$.

- The probability p_1 that a value we condition upon is bad. This error is incurred by the condenser and is high (think of it as being a constant).
- We show that when we condition on a good value, every t good rows in Y''' are p_2-close to uniform. We then claim that Y''' as a table is $A^t p_2$-close to a table where the good rows are truly t-wise independent (where A is the number of rows in the table Y'''). The error p_2 is incurred by Raz's extractor and by the correlation-breaker, and can be made very small if we deal with a source X having enough min-entropy. We make p_2 small enough so that $A^t p_2$ is also small.
- A third error p_3 is incurred by the resilient function f. This error is large, say, a constant, and we are fine with that.

Note that we cannot just accumulate all errors as $A^t p_1$ is way larger than 1. Instead, we argue that with a constant probability $1-p_1$, we get extremely close to perfect behavior, and then we get such a small error p_2 so that $A^t p_2$ is also small.

4.3 The Analysis

Proof of Theorem 27. Fix an (n,k) source X and an independent $(d,\delta d)$ source Y. We decompose the proof into three parts:

- In the first part we prove that very often (except for a small constant probability) the table Y'' contains many rows that are marginally close to uniform.
- Next, we prove that every set of t rows $\{i,j_1,\ldots,j_{t-1}\}$ in Y''' are product in the sense that if i is a good row (intuitively meaning that Y''_i is close to uniform) and $j_1,\ldots,j_{t-1}$ are $t-1$ other rows, then in Y''', Y'''_i is close to uniform and independent of $Y'''_{j_1},\ldots,Y'''_{j_{t-1}}$. This part involves applying a correlation-breaker with advice on Y and Y''. In order to ensure that Y and Y'' are independent, we condition on the values of Y' in the t rows $\{i,j_1,\ldots,j_{t-1}\}$.
- Together, except for a small constant probability, there are many good rows, and every t rows of Y''' are product, hence the table Y''' is close to a (q,t) non-oblivious bit-fixing source, where every good row is a good bit in the bit-fixing source. Hence, $f(Y''')$ is close to uniform.

Part 1: Often, many rows in Y' are good. Let ε_i be the minimal distance of $C(Y,i)$ from a $\delta' d'$-source. According to Lemma 9,

$$\mathbb{E}_{i\in[A]}[\varepsilon_i] \leq \varepsilon_{\text{cond}}.$$

Definition 28. *We say $z\in\{0,1\}^{d'}$ is good if $\text{Raz}(X,z)$ is ε_{Raz}-close to uniform. Let G_Z be the set of all good z-s, and B_Z the rest. We say $i\in[A]$ is good for $y\in\{0,1\}^d$ if $C(y,i)\in G_Z$ and bad otherwise. We define a random variable B_i, where the sample space is Y, and $B_i(y)=1$ if i is bad for y and 0 otherwise.*

By Claim 12, $|B_Z|\leq 2^{\delta_{\text{Raz}}d'}$. Therefore, in expectation, the number of bad rows for y is small:

Claim 29. $\mathbb{E}_{y\in Y}[\sum_{i\in[A]} B_i(y)] \le 2\varepsilon_{\text{cond}} A$.

Proof. Fix an $i \in [A]$. We have that $C(Y,i)$ is ε_i-close to some $\delta' d' = 0.69d'$-source R. Hence:

$$\mathbb{E}_y[B_i(y)] = \Pr_{y\in Y}[C(y,i) \in B_Z] \le \varepsilon_i + \Pr_{r\in R}[r \in B_Z] \le \varepsilon_i + \frac{|B_Z|}{2^{\delta' d'}} = \varepsilon_i + 2^{-0.09d'}.$$

Thus, for d large enough,

$$\begin{aligned}\mathbb{E}_y\left[\sum_{i\in[A]} B_i(y)\right] &= \sum_{i\in[A]} \mathbb{E}_y[B_i(y)] \le \sum_{i\in[A]} (\varepsilon_i + 2^{-0.09d'}) \\ &\le \varepsilon_{\text{cond}} A + 2^{-0.09d'} A \le 2\varepsilon_{\text{cond}} A.\end{aligned}$$

□

Definition 30. *We say $y \in \text{Supp}(Y)$ has many bad rows if $\sum_{i\in[A]} B_i(y) \ge \sqrt{\varepsilon_{\text{cond}}} A$.*

Denote $p_{1,1} = \varepsilon_0/4$.

Claim 31. $\Pr_{y\in Y}[y \text{ has many bad rows}] \le p_{1,1}$.

Proof. By Markov,

$$\begin{aligned}\Pr_{y\in Y}\left[\sum_i B_i(y) \ge \sqrt{\varepsilon_{\text{cond}}} A\right] &\le \frac{\mathbb{E}[\sum_i B_i(y)]}{\sqrt{\varepsilon_{\text{cond}}} A} \\ &\le \frac{2\varepsilon_{\text{cond}} A}{\sqrt{\varepsilon_{\text{cond}}} A} = 2\sqrt{\varepsilon_{\text{cond}}} \le \frac{\varepsilon_0}{4},\end{aligned}$$

where the last inequality follows from the fact that $\varepsilon_{\text{cond}} \le (\varepsilon_0/8)^2$. □

Part 2: The good rows are t-wise independent. We introduce some notations to simplify the expressions in the proof. For $y_0 \in \{0,1\}^d$ and $k \in [A]$, let $Y'''_k(y_0)$ denote $(Y'''_k \mid Y = y_0)$. Also, for a set $S \subseteq [A]$, define $Y'''_S(y_0) = \{Y'''_j(y_0)\}_{j\in S}$. Denote $p_2 = \varepsilon_{\text{Raz}} + 2\varepsilon_{\text{CBA}}$.

Definition 32. *Let $y_0 \in \{0,1\}^d$ (not necessarily in the support of Y). Let $i \in [A]$ and $S \subseteq [A] \setminus \{i\}$ of cardinality $t-1$. We say y_0 violates the product rule for (i,S) if $B_i(y_0) = 0$ and*

$$(Y'''_i(y_0), Y'''_S(y_0)) \not\approx_{p_2} U_1 \times Y'''_S(y_0).$$

Definition 33. *Let $y_0 \in \{0,1\}^d$ (not necessarily in the support of Y). Let $i \in [A]$ and $S \subseteq [A] \setminus \{i\}$ of cardinality $t-1$. We say y_0 violates the product rule with distinguisher $\Delta : \{0,1\}^t \to \{0,1\}$ for (i,S) if $B_i(y_0) = 0$ and*

$$|\Pr[\Delta(Y'''_i(y_0), Y'''_S(y_0)) = 1] - \Pr[\Delta(U_1, Y'''_S(y_0)) = 1]| > p_2.$$

Observe that if y_0 violates the product rule then there exists some Δ such that y_0 violates the product rule with distinguisher Δ.

Lemma 34. *For every i and S as above, the number of $y \in \{0,1\}^d$ that violate the product rule for (i,S) is at most $2^{\delta_0 d/2+2^t}$.*[2]

Proof. Suppose the lemma is false for some (i,S). Then, by the pigeonhole principle there exists some Δ such that the number of elements $y \in \{0,1\}^d$ that violate the product rule for (i,S) with distinguisher Δ is at least $2^{\delta_0 d/2}$. Let BY denote the set of these elements. Identify BY with the uniform distribution over the set BY. Let $BY_i' = C(BY,i)$, $BY_i'' = \mathrm{Raz}(X, BY_i')$ and $BY_i''' = \mathrm{CBA}(BY, BY_i'', i)$. For a subset $T \subseteq [A]$ Let BY_T' denote the sub-table of BY' corresponding to the rows of T, and similarly BY_T'' and BY_T'''. Since for every $y \in BY$, we have that

$$\Delta(BY_i'''(y), BY_S'''(y)) \not\approx_{p_2} \Delta(U_1, BY_S'''(y)),$$

this holds also on average, that is

$$\Delta(BY_i''', BY_S''') \not\approx_{p_2} \Delta(U_1, BY_S'''').$$

Thus, it follows that

$$BY'''_{S\cup\{i\}} \not\approx_{p_2} U_1 \times BY_S'''. \tag{3}$$

On the other hand, when we condition on the values of $H = BY'_{S\cup\{i\}}$, the conditions for the correlation-breaker with advice hold:

- BY and $BY''_{S\cup\{i\}}$ are independent given $H = BY'_{S\cup\{i\}}$, since H is a function of BY alone, and given that $H = BY'_{S\cup\{i\}} = h$ for some h, $BY''_{S\cup\{i\}}$ is a function of X alone.
- It holds that

  $$\begin{aligned}\widetilde{H}_\infty(BY \mid H) &\geq H_\infty(BY) - \log(|\operatorname{Supp}(H)|) \\ &= H_\infty(BY) - td' \geq \frac{\delta_0 d}{2} - td' \geq \frac{\delta_0 d}{4},\end{aligned}$$

  because

  $$\frac{td'}{d} = t \cdot \left(\frac{2}{3}\right)^s = \frac{\delta_0}{4}.$$

  Now, since $k_{\mathrm{CBA}} = \frac{\delta_0 d}{8}$ and $\varepsilon_{\mathrm{CBA}} = \frac{1}{d}$ we also have for d large enough,

  $$\widetilde{H}_\infty(BY \mid H) \geq \frac{\delta_0 d}{4} \geq k_{\mathrm{CBA}} + \log\frac{1}{\varepsilon_{\mathrm{CBA}}}.$$

- $B_i(y) = 0$, hence $BY_i' \in G_Z$ and $BY_i'' = \mathrm{Raz}(X, BY_i')$ is $\varepsilon_{\mathrm{Raz}}$-close to uniform.

Thus, by the correlation-breaker with advice property,

$$\begin{aligned}&(\mathrm{CBA}(BY, BY_i'', i), \{\mathrm{CBA}(BY, BY_j'', j)\}_{j\in S}) \\ &\qquad\qquad \approx_{\varepsilon_{\mathrm{Raz}}+2\varepsilon_{\mathrm{CBA}}} (U_1, \{\mathrm{CBA}(BY, BY_j'', j)\}_{j\in S}),\end{aligned}$$

[2] We could have used an alternative argument that avoids the 2^{2^t} factor here by a minor deterioration in the error of the CBA. However, since the t we use is constant the 2^{2^t} factor is negligible.

or, equivalently,

$$(BY_i''', BY_S''') \approx_{p_2} U_1 \times BY_S''',$$

in contradiction to Equation (3). □

Definition 35. *Say $y \in \{0,1\}^d$ violates the product rule if it violates it for some $i \in [A]$ and $S \subseteq [A] \setminus \{i\}$ of cardinality $t-1$.*

As $H_\infty(Y) \geq \delta_0 d$, the probability $y \in Y$ violates the product rule for a specific (i, S) is at most $2^{\delta_0 d/2 + 2^t - \delta_0 d} = 2^{2^t - \delta_0 d/2}$. Let $p_{1,2} = \varepsilon_0/10$. Then, by the union bound, for d large enough:

Corollary 36. $\Pr_{y \in Y}[y \text{ violates the product rule}] \leq 2^{2^t - \delta_0 d/2} \cdot A^t \leq p_{1,2}$.

Part 3: Completing the proof.

Definition 37. *We say y is bad if it has many bad rows or if it violates the product rule. If y is not bad we say it is good.*

Let $p_1 = p_{1,1} + p_{1,2}$. Clearly, by Claim 31 and Corollary 36, $\Pr_{y \in Y}[y \text{ is bad}] \leq p_1 = (\frac{1}{4} + \frac{1}{10})\varepsilon_0$.

Claim 38. *Fix any good $y \in Y$. Then, $Y'''(y)$ is a (q, t, tp_2) non-oblivious bit-fixing source, for $q = \sqrt{\varepsilon_{\text{cond}}} A$.*

Proof. Let $Q(y) \subseteq [A]$ be the set of bad rows for y. As y does not have many bad rows, $|Q(y)| = \sum_{i \in [A]} B_i(y) \leq \sqrt{\varepsilon_{\text{cond}}} A = q$. Now, fix any set $S \subseteq [A] \setminus Q(y)$ of cardinality t. Let $i \in S$. As $S \subseteq [A] \setminus Q(y)$ and $i \in S$ we have $i \notin Q(y)$ and therefore $B_i(y) = 0$. Also, y does not violate the product rule, hence,

$$(Y_i'''(y), Y'''_{S \setminus \{i\}}(y)) \approx_{p_2} U_1 \times Y'''_{S \setminus \{i\}}(y).$$

As this is true for any $i \in S$, by Lemma 18,

$$Y_S'''(y) \approx_{tp_2} U_t.$$

Thus, $Y'''(y)$ is a (q, t, tp_2) non-oblivious bit-fixing source. □

In particular, by Lemma 20, for every good y, $Y'''(y)$ is $tA^t p_2$-close to a (q, t) non-oblivious bit-fixing source. By the choices we have made above $q = \sqrt{\varepsilon_{\text{cond}}} A \leq A^{1-2\gamma}$. Equation (1) implies that

$$t = \frac{\delta_0}{4} \left(\frac{1}{\varepsilon_{\text{cond}}} \right)^{\log(3/2)} \geq c_\gamma \log^4 A.$$

Using the resiliency of f from Theorem 22 (and the fact that it is almost balanced), the output when y is good is p_3-close to uniform for $p_3 = tA^t p_2 + \varepsilon_f + A^{-1/c_\gamma}$, where the first term is due to the distance from a t-wise distribution, the second is due to the resiliency and the third is due to the bias of f (see, e.g., Lemma 2.11 in [9]). To that we also have to add the probability p_1 that y is not good. To finish the proof we notice that:

- It holds that

$$\varepsilon_f \leq c_\gamma \frac{q}{A^{1-\gamma}} \leq c_\gamma \frac{A^{1-2\gamma}}{A^{1-\gamma}} = c_\gamma A^{-\gamma}$$
$$\leq c_\gamma 2^{-\gamma \log \frac{1}{\varepsilon_{\text{cond}}}} \leq \frac{\varepsilon_0}{5},$$

  because $\varepsilon_{\text{cond}} \leq (\frac{\varepsilon_0}{5c_\gamma})^{1/\gamma}$.

- Also,

$$A^{-1/c_\gamma} \leq 2^{-\frac{1}{c_\gamma} \log \frac{1}{\varepsilon_{\text{cond}}}} = \varepsilon_{\text{cond}}^{1/c_\gamma} \leq \frac{\varepsilon_0}{5},$$

  because $\varepsilon_{\text{cond}} \leq (\frac{\varepsilon_0}{5})^{c_\gamma}$.

- Finally, $tp_2 = t(\varepsilon_{\text{Raz}} + 2\varepsilon_{\text{CBA}})$, $\varepsilon_{\text{Raz}} = 2^{-\Omega(k)} = d^{-\Omega(1)}$, $\varepsilon_{\text{CBA}} = \frac{1}{d}$. Thus, $tp_2 \leq 4td^{-\Omega(1)}$. A and t are constants, so for d large enough, $tA^t p_2 \leq \frac{\varepsilon_0}{5}$.

Together, the error is at most $p_1 + p_3 \leq \varepsilon_0$ completing the proof of the theorem. □

Strong Seeded Dispersers and Friends (original Sections 5.1 and 5.3)

5.1 Strong Seeded Dispersers

Definition 39 (strong disperser)**.** *A function* $\text{Disp} : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ *is a strong* (k, ε) *disperser, if for every* (n, k) *source* X*,*

$$|\operatorname{Supp}((Y, \text{Disp}(X, Y)))| > (1 - \varepsilon)DM.$$

We say Disp *is (source) linear if for every* $y \in \{0,1\}^d$ *and every* $x_1, x_2 \in \mathbb{F}_2^n$*,* $\text{Disp}(x_1 + x_2, y) = \text{Disp}(x_1, y) + \text{Disp}(x_2, y)$*.*

We are interested in the important special case where $m = 1$. In this case, non-explicitly, a random function is (w.h.p.) a strong (k, ε) disperser with $d = \log n + \log(\frac{1}{\varepsilon}) + O(1)$ provided that $k \geq \log\log(\frac{1}{\varepsilon}) + O(1)$ [37,33]. A matching lower bound, up to additive constant factors, was given by [37]. Using the translation between strong seeded dispersers and erasure list-decodable codes which we discuss in Section 5.3, Guruswami and Indyk's result [26] gives a probabilistic polynomial time algorithm that outputs with high probability a strong seeded disperser with seed-length $d = 2\log(\frac{1}{\varepsilon}) + \log n + \log\log(\frac{1}{\varepsilon})$ and optimal entropy-loss. The construction can be made deterministic, but with running time exponential in $1/\varepsilon$. Note that as we discuss the one output bit case, the required entropy is essentially the entropy-loss. From Theorem 27 we can derive a better explicit construction of a strong disperser with small error.

Theorem 40. *For every constant* $0 < \gamma < 1$ *there exists a constant* $c \geq 1$ *such that for every integer* n *and* $\varepsilon \leq n^{-\frac{c}{1-\gamma}}$ *there exists an explicit strong* (k, ε) *disperser* $\text{Disp} : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}$ *where* $d = (1 + \gamma)\log(\frac{1}{\varepsilon})$ *and* $k = c \log d$*.*

Proof. Set $\varepsilon_0 = \frac{1}{4}$ and $\delta_0 = \frac{\gamma}{1+\gamma}$. Let c be the constant from Theorem 27 for δ_0 and ε_0 and let $\mathsf{2Ext} : [N] \times [D] \to \{0,1\}$ be the $((n,k),(d,k_2 = \delta_0 d),\varepsilon_0)$ two-source extractor where $d = (1+\gamma)\log(\frac{1}{\varepsilon})$ and $k = c\log d$. Notice that $d \geq c\log n$ (because $\varepsilon \leq n^{-\frac{c}{1+\gamma}}$) as required. Let $\mathrm{Disp}(x,y) = \mathsf{2Ext}(x,y)$. Let $X \subseteq [N]$ be a set of size K and call a value $y \in [D]$ b-bad if $\mathrm{Disp}(X,y) = \{b\}$. It follows that the sets of 0-bad y-s and 1-bad y-s are each of size less than K_2. Therefore,

$$\begin{aligned} |\operatorname{Supp}((U_d, \mathrm{Disp}(X,U_d)))| &> 2K_2 + 2(D - 2K_2) = 2D - 2K_2 \\ &= \left(1 - \frac{K_2}{D}\right) 2D = (1-\varepsilon)2D, \end{aligned}$$

because $\frac{K_2}{D} = 2^{-(1-\delta_0)d} = 2^{-\frac{1}{1+\gamma}d} = 2^{-\log(\frac{1}{\varepsilon})} = \varepsilon$. □

5.3 Erasure List-Decodable Codes

An $(\bar{n}, n)$ (binary) code is a mapping $\mathcal{C} : \{0,1\}^n \to \{0,1\}^{\bar{n}}$. The code $\mathcal{C}$ is linear if $\mathcal{C}$ is linear, and is denoted by $[\bar{n}, n]$. We identify a code with the image of $\mathcal{C}$. For a linear $\mathcal{C}$ this image is a linear subspace of $\mathbb{F}_2^{\bar{n}}$ of dimension n. A generator matrix for an $[\bar{n}, n]$ code $\mathcal{C}$ is any matrix whose columns form a basis for $\mathcal{C}$. In the erasures noise model, an adversarially chosen subset of the codeword's symbols are erased and the positions where erasures have occurred are known.

Definition 41 (erasure list-decodable code)**.** *A code $\mathcal{C} \subseteq \{0,1\}^{\bar{n}}$ is (s, L) erasure list-decodable if for every $r \in \{0,1\}^{\bar{n}-s}$ and every set $T \subseteq [\bar{n}]$ of size $\bar{n} - s$,*

$$|\{c \in \mathcal{C} \mid c|_T = r\}| < L,$$

where $c|_T$ denotes the projection of c to the coordinates in T.

The following folklore lemma (see, e.g., [24, Lemma 1]) gives an alternative characterization of linear erasure list-decodable codes.

Lemma 42. *An $[\bar{n}, n]_2$ linear code $\mathcal{C}$ is $((1-\varepsilon)\bar{n}, L)$ erasure list-decodable if and only if its $\bar{n} \times n$ generator matrix G has the property that every $\varepsilon\bar{n} \times n$ sub-matrix of G has rank greater than $n - \log L$.*

Non-explicitly, we have:

Theorem 43 ([24])**.** *For every n and $\varepsilon > 0$, there exists an $(\bar{n}, n)$ binary code that is $((1-\varepsilon)\bar{n}, L)$-erasure list-decodable of rate $\frac{n}{\bar{n}} = \Omega(\varepsilon)$ and $L = O(\log(1/\varepsilon))$.*

Guruswami [25] observed that strong dispersers can be used to construct erasure list-decodable codes. Here we complement his argument, and note that strong dispersers are equivalent to erasure list-decodable codes. Given a function $\mathrm{Disp} : [N] \times [D] \to \{0,1\}$, we consider the (D, n) code $\mathcal{C}_{\mathrm{Disp}} : \{0,1\}^n \to \{0,1\}^D$ defined by $\mathcal{C}_{\mathrm{Disp}}(x)_i = \mathrm{Disp}(x,i)$. Note that the code is linear if and only if Disp is linear.

Lemma 44 (following [25, Lemma 12])**.** *The function $\mathrm{Disp} : [N] \times [D] \to \{0,1\}$ is a strong (k,ε) disperser if and only if $\mathcal{C}_{\mathrm{Disp}}$ is $((1-2\varepsilon)D, K)$ erasure list-decodable.*

Proof. For one direction, assume Disp is a strong (k, ε) disperser. We wish to prove that $\mathcal{C}_{\mathrm{Disp}}$ is $((1-2\varepsilon)D, K)$ erasure list-decodable. Let $T = \{t_1, \ldots, t_{2\varepsilon D}\} \subseteq [D]$ be an arbitrary set of size $2\varepsilon D$ and $r \in \{0,1\}^{2\varepsilon D}$ an arbitrary string. Let $X_{T,r} \subseteq \{0,1\}^n$ denote the set of all the messages x for which $\mathcal{C}_{\mathrm{Disp}}(x)|_T = r$. Then,

$$|\operatorname{Supp}((U_d, \operatorname{Disp}(X_{T,r}, U_d)))| \leq |T| \cdot 1 + (D - |T|) \cdot 2 \leq (1-\varepsilon)2D,$$

where the first inequality follows by considering seeds in T and seeds in $[D] \setminus T$. For a seed $t_i \in T$ we have that $\operatorname{Disp}(X_{T,r}, t_i)$ is fixed, hence each such seed contributes 1 to the support size. For any other seed y, the support size of $\operatorname{Disp}(X_{T,r}, y)$ is at most 2. As Disp is a strong (k, ε) disperser, we conclude that $|X_{T,r}| \leq K$ as desired. For the other direction assume Disp is a not a strong (k, ε) disperser. Then, there exists a set $X \subseteq \{0,1\}^n$ such that $|X| \geq K$ and $|\operatorname{Supp}((U_d, \operatorname{Disp}(X, U_d)))| \leq (1-2\varepsilon)2D$. Note that for every $y \in [D]$ we have $|\operatorname{Supp}(\operatorname{Disp}(X, y))| \in \{1, 2\}$. Therefore, following the above calculation, there exists a set $T \subseteq D$ of size at least $2\varepsilon D$ such that for each $y \in T$, $|\operatorname{Supp}(\operatorname{Disp}(X, y))| = 1$. But this means that for every $x \in X$, $\mathcal{C}_{\mathrm{Disp}}(x)|_T$ is the same (punctured) codeword. It follows that $\mathcal{C}_{\mathrm{Disp}}$ is not $((1-2\varepsilon)D, K)$ erasure list-decodable. □

Corollary 45. *If* $\mathrm{Disp} : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}$ *is a strong* (k, ε) *disperser with seed-length* $d = a_1 \log n + a_2 \log(\frac{1}{\varepsilon}) + a_3$ *(for some* $a_1 \geq 1, a_2 \geq 1$ *and* a_3*) then* $\mathcal{C}_{\mathrm{Disp}}$ *is a* $((1-2\varepsilon)D, K)$ *erasure list-decodable code of rate* $2^{-a_3} \cdot n^{1-a_1} \cdot \varepsilon^{a_2}$.

When ε is much smaller than $\frac{1}{n}$ the dominant factor is determined by a_2. As we mentioned earlier (and as Guruswami also notes in [25]) previous explicit constructions for binary codes had $a_2 \geq 2$ (usually inherited from extractor constructions). Our construction is the first to get arbitrary close to $a_2 = 1$ and small list-size. Combining Corollary 45 and Theorem 40, we obtain:

Theorem 46. *For every constant* $0 < \gamma < 1$ *there exists a constant* $c \geq 1$ *such that for every integer* n *and* $\varepsilon \leq n^{-\frac{c}{1-\gamma}}$ *there exists an explicit code* $\mathcal{C} : \{0,1\}^n \to \{0,1\}^{(\frac{1}{\varepsilon})^{1+\gamma}}$ *that is*

$$\left((1-2\varepsilon)\left(\frac{1}{\varepsilon}\right)^{1+\gamma}, \left((1+\gamma)\log\frac{1}{\varepsilon}\right)^c\right)$$

erasure list-decodable of rate $n\varepsilon^{1+\gamma}$.

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